

# CS Capstone Design

## Technical Demo Grading Sheet (100 pts)

### TEAM: KIMinder360 Team

**Overview:** The main purpose of the “Technical Demos” is to very clearly communicate the extent to which the team has identified key challenges in the project, and has proven solutions to those challenges. Grading is based on how complete/accurate the list of challenges is, , and how convincingly and completely the given demos cover the given challenges.

This template is fleshed out by the team, approved by CS mentor, and brought to demo as a grading sheet.

### Risky technical challenges

Based on our requirements acquisition work and current understanding of the problem and envisioned solution, the following are the key technical challenges that we will need to overcome in implementing our solution:

**C1: Correctly scheduling time-bounded notifications.** Active hours must be customizable by the user to allow them to select when they receive directive notifications. User preferences should for start and stop times should be stored to have time boundaries for when notifications can be sent out.

**C2: Designing a scalable many-to-many schema in Room.** When implementing our persistent data storage we need to link every single directive to one or more directive categories they could be a part of. This gives our application its core functionality by having a collection of directives to pull from for notifications as well as categorization of them into subsets. Without this our app is unable to have a collection of unique prompts for the user.

**C3: Push Notifications.** We need to be able to set specific times for notifications to be sent out and send them to the user easily. If our app is unable to send the user notifications at set times then we cannot deliver directives. Android aggressively limits background tasks so we must have a way to ensure notifications fire at exact times.

**C4: Category Selection.** We must allow the user to select which categories they want to receive directives from. If the user changes their selected categories, previously scheduled notifications may become invalid. The app must dynamically adjust the plan without causing skipped notifications or dispatching prompts from disabled categories.

### Challenges covered by demos:

In this section, we outline the demonstrations we have prepared, and exactly which of the challenge(s) each one of them proves a solution to.

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#### Demonstration 1: View and Customize Active Hours

Challenges addressed: C1

Flight Plan: Step by step overview of demo

1. First open the app

2. Then click on the customize active hours button
3. Then activate the monday hours
4. Then Set the monday hours to 9 to 5
5. Then select the cancel button to exit active hours screen
6. The select active hours customization
7. Then turn on active hours for the current day and 2 other days
8. Then click save to confirm changes and exit the screen
9. Finally, view the active hours being displayed on the home screen of the app

Evaluation:

- ✓ Convincingly demo'd each of listed challenges?
- ✓ Other evaluative comments:

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**Demonstration 2: View and customize directive category selection**

Challenges addressed: C4

Flight Plan: Step by step overview of demo

1. First click on the select categories button
2. Then uncheck 3 of the categories
3. Then select the cancel button to discard your changes
4. Select the categories button
5. Uncheck all categories then reselect 4 of them
6. Select the save button to confirm your changes
7. Select the categories button to show that the changes are saved

Evaluation:

- ✓ Convincingly demo'd each of listed challenges?
- ✓ Other evaluative comments:

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**Demonstration 3: View notifications from KIMinder360**

Challenges addressed: C3

Flight Plan: Step by step overview of demo

1. First close the app
2. Then pull down the notification shade
3. Then click on the KIMinder360 Notifications to view all of them
4. Finally click on a KIMinder360 Notification to open the app

Evaluation:

✓ Convincingly demo'd each of listed challenges?

✓ Other evaluative comments:

**Other challenges recognized but not addressed by demo:**

If there were challenges you listed earlier that were *not* covered by a demo, list here. This will hopefully be a short list...but better to be clear about where you are. If you have items here, you could list (if applicable) any pending plans to reduce these risks.

C2: Room Database - The database will be implemented, but not initially planned for demonstration via the category selection or push notifications.